

Lab Program 5

Title: Demonstrate CI Integration by running Jenkins in a Docker container

File Structure:

prog5/

 docker-compose.yml

 Dockerfile

STEP 1: Check if docker and docker compose are installed

\$ docker -v

\$ docker compose version

STEP 2: Create program directory

\$ mkdir lab5 && cd lab5

STEP 3: Create Dockerfile

\$ nano Dockerfile

```
FROM jenkins/jenkins:lts

USER root

RUN apt-get update && apt-get install -y docker.io

USER jenkins
```

STEP 3: Create Docker Compose file

\$ nano docker-compose.yml

```
services:
  jenkins:
    build: .
    container_name: jenkins-ci
    ports:
      - "8080:8080"
      - "50000:50000"
    volumes:
      - jenkins_home:/var/jenkins_home
      - /var/run/docker.sock:/var/run/docker.sock

volumes:
  jenkins_home:
```

STEP 4: Build and Run Jenkins

Using Docker Compose

\$ docker compose up -d --build

```
victus% docker compose up -d --build

[+] Building 52.3s (9/9) FINISHED
  => [internal] load local bake definitions                                0.0s
  => => reading from stdin 556B                                           0.0s
  => [internal] load build definition from Dockerfile                     0.0s
  => => transferring dockerfile: 139B                                       0.0s
  => [internal] load metadata for docker.io/jenkins/jenkins:lts          2.5s
  => [auth] jenkins/jenkins:pull token for registry-1.docker.io          0.0s
  => [internal] load .dockerignore                                         0.0s
  => => transferring context: 2B                                           0.0s
  => CACHED [1/2] FROM docker.io/jenkins/jenkins:lts@sha256:1e407df1a9bcb122cc4d71d4ce44 0.0s
  => [2/2] RUN apt-get update && apt-get install -y docker.io            48.6s
  => exporting to image                                                    1.0s
  => => exporting layers                                                    0.9s
  => => writing image sha256:4e692a6538c2897d7f035f37cfef73614475e8795a579fcc60031c6c0f9 0.0s
  => => naming to docker.io/library/lab5-jenkins                          0.0s
  => resolving provenance for metadata file                                0.0s
[+] up 2/2
  ✓ Image lab5-jenkins      Built                                         52.3s
  ✓ Container jenkins-ci    Recreated                                      0.1s
```

STEP 5: Test the running of Jenkins by logging in to

<http://localhost:8080>



Sign in to Jenkins

Username

Password

☐ Keep me signed in

Sign in

STEP 6: Check running containers

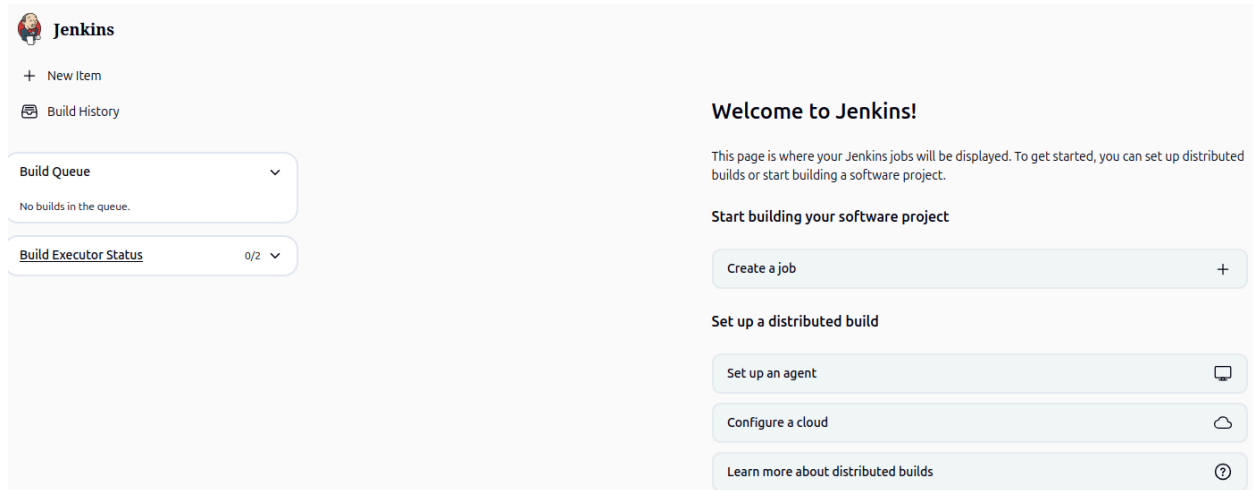
\$ docker compose ps

```
victus% docker compose ps
NAME                IMAGE             COMMAND                                     SERVICE  CREATED        STATUS      PORTS
jenkins-ci          lab5-jenkins      "/usr/bin/tini -- /u..."               jenkins  17 minutes ago Up 17 minutes 0.0.0.0:8080→8080/tcp,
[::]:8080→8080/tcp, 0.0.0.0:50000→50000/tcp, [::]:50000→50000/tcp
```

STEP 7: Set up username and password if username chosen as “*admin*”
Get the password by running

```
$ docker exec -it jenkins-ci cat /var/jenkins_home/secrets/initialAdminPassword
```

```
victus% docker exec -it jenkins-ci cat /var/jenkins_home/secrets/initialAdminPassword  
#11a3b84442f4d58b4decf8caf521430
```



STEP 8: Stop and delete the container

```
$ docker stop jenkins-ci
```

```
$ docker rm jenkins-ci
```

```
victus% docker stop jenkins-ci  
jenkins-ci  
victus% docker rm jenkins-ci  
jenkins-ci
```

STEP 9: Delete the docker image

```
$ docker image rm lab5-jenkins:latest
```

```
victus% docker image rm lab5-jenkins:latest  
Untagged: lab5-jenkins:latest  
Deleted: sha256:4e692a6538c2897d7f035f37cfeb73614475e8795a579fcc60031c6c0f9bb8ad
```

Using Dockerfile

\$ sudo docker build -t prog5 .

```
victus% sudo docker build -t prog5 .
[+] Building 2.6s (7/7) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 139B                             0.0s
=> [internal] load metadata for docker.io/jenkins/jenkins:lts  2.4s
=> [auth] jenkins/jenkins:pull token for registry-1.docker.io  0.0s
=> [internal] load .dockerignore                                0.0s
=> => transferring context: 2B                                    0.0s
=> [1/2] FROM docker.io/jenkins/jenkins:lts@sha256:1e407df1a9bcb122cc4d71d4ce44d2e3b2b 0.0s
=> CACHED [2/2] RUN apt-get update && apt-get install -y docker.io 0.0s
=> exporting to image                                           0.0s
=> => exporting layers                                           0.0s
=> => writing image sha256:2a8239b38d17d4f1769d965a8a6861c6f3cff2d1b71d909401a09cf5d89 0.0s
=> => naming to docker.io/library/prog5                          0.0s
```

STEP 10: Run the container

\$ docker run -p 8080:8080 -p 50000:50000 --name jenkins-ci prog5

```
victus% docker run -p 8080:8080 -p 50000:50000 --name jenkins-ci prog5
Running from: /usr/share/jenkins/jenkins.war
webroot: /var/jenkins_home/war
2026-01-02 05:59:11.021+0000 [id=1]      INFO      winstone.Logger#logInternal: Beginn
ing extraction from war file
```

STEP 11: Test the running of Jenkins by logging in to <http://localhost:8080>

Getting Started

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log ([not sure where to find it?](#)) and this file on the server:

```
/var/jenkins_home/secrets/initialAdminPassword
```

Please copy the password from either location and paste it below.

Administrator password

Continue

Paste the password generated for the admin

STEP 12: Stop and delete containers

```
victus% docker rm a172ef858980  
a172ef858980
```

```
victus% docker image rm prog5:latest  
Untagged: prog5:latest  
Deleted: sha256:2a8239b38d17d4f1769d965a8a6861c6f3cff2d1b71d909401a09cf5d89d71a2
```